Distilling MACE-Polar into a fast linear student

Architecture, six-molecule accuracy, and speed of the charge/dipole distillation

27 October 2026 · Symmetrix-Polar / Plan D

KEY NUMBERS

E / F, 3,465 held-out frames

1.2 / 7.8

meV atom^{-1} / meV A^{-1}

Six-molecule gas

1.17 / 1.00

meV atom^{-1} / meV A^{-1}

Atomic charge q

0.0117 e

867 frames · 4π scale

l=1 dipole μ

0.0475

e A · 4π scale

GTO/Ewald from q, μ

0.311

meV atom^{-1} overall

Speed, 5,184 atoms

4.47 / ~20.2

backbone-only FP32 / full FP64 (proj.)

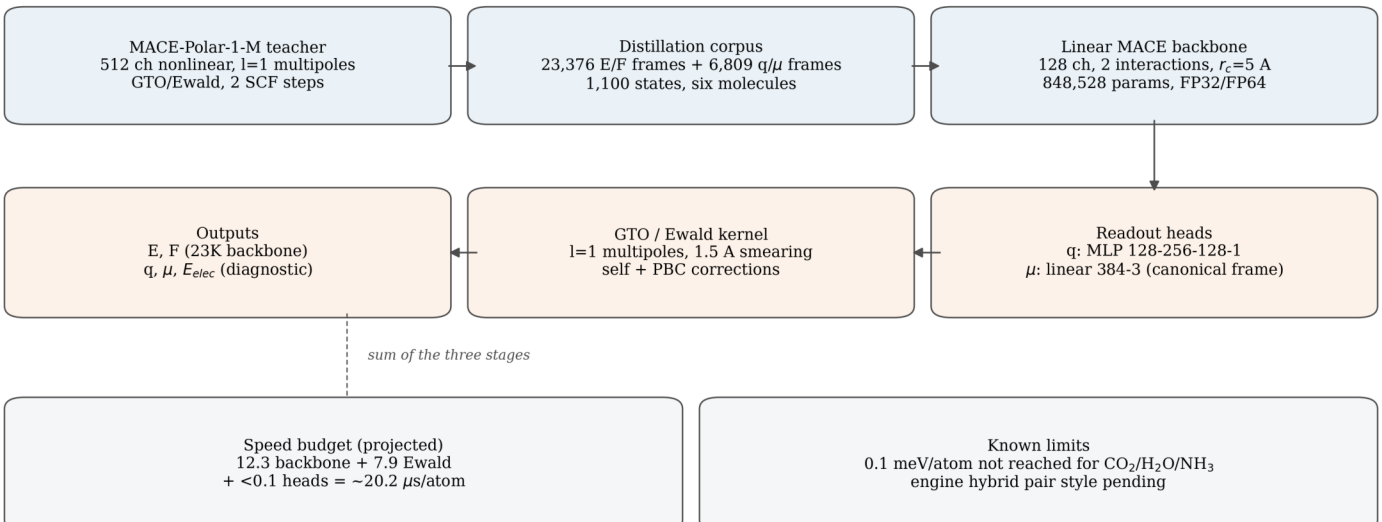
A 128-channel linear MACE backbone distilled from the MACE-Polar-1-M teacher reaches 1.2 meV/atom and 7.8 meV/A on the full 23,376-frame corpus (3,465 held-out). A charge MLP and a linear l=1 multipole readout, trained on 6,809 frames covering all 1,100 states, predict q to 0.0117 e and mu to 0.0475 (4pi scale). Feeding the predicted multipoles through the teacher's GTO/Ewald kernel reproduces the teacher electrostatic energy to 0.311 meV/atom overall.

The student is stable in NVE (0.245 meV/atom/ps over 1 ps), its forces rotate to 4.0e-5, and a canonical-frame transform makes the full linear head rotation-consistent to 4.6e-14 meV/atom (0.825 without it). The projected speed is ~20.2 us/atom/step: 12.3 engine backbone + 7.9 PyTorch Ewald + <0.1 heads, with a 1.2x k-space cutoff giving <=20 us. An end-to-end engine hybrid pair style is the remaining implementation.

Scope. The q/mu readout is trained on 6,809 frames, not the full 23K; the remaining frames are mostly correlated frames inside the same 1,100 states.

CO2/H2O/NH3 retain 0.4-0.6 meV/atom Ewald error; 0.1 meV/atom is not claimed. The ~20 us speed is a projection, not a measured end-to-end engine run.

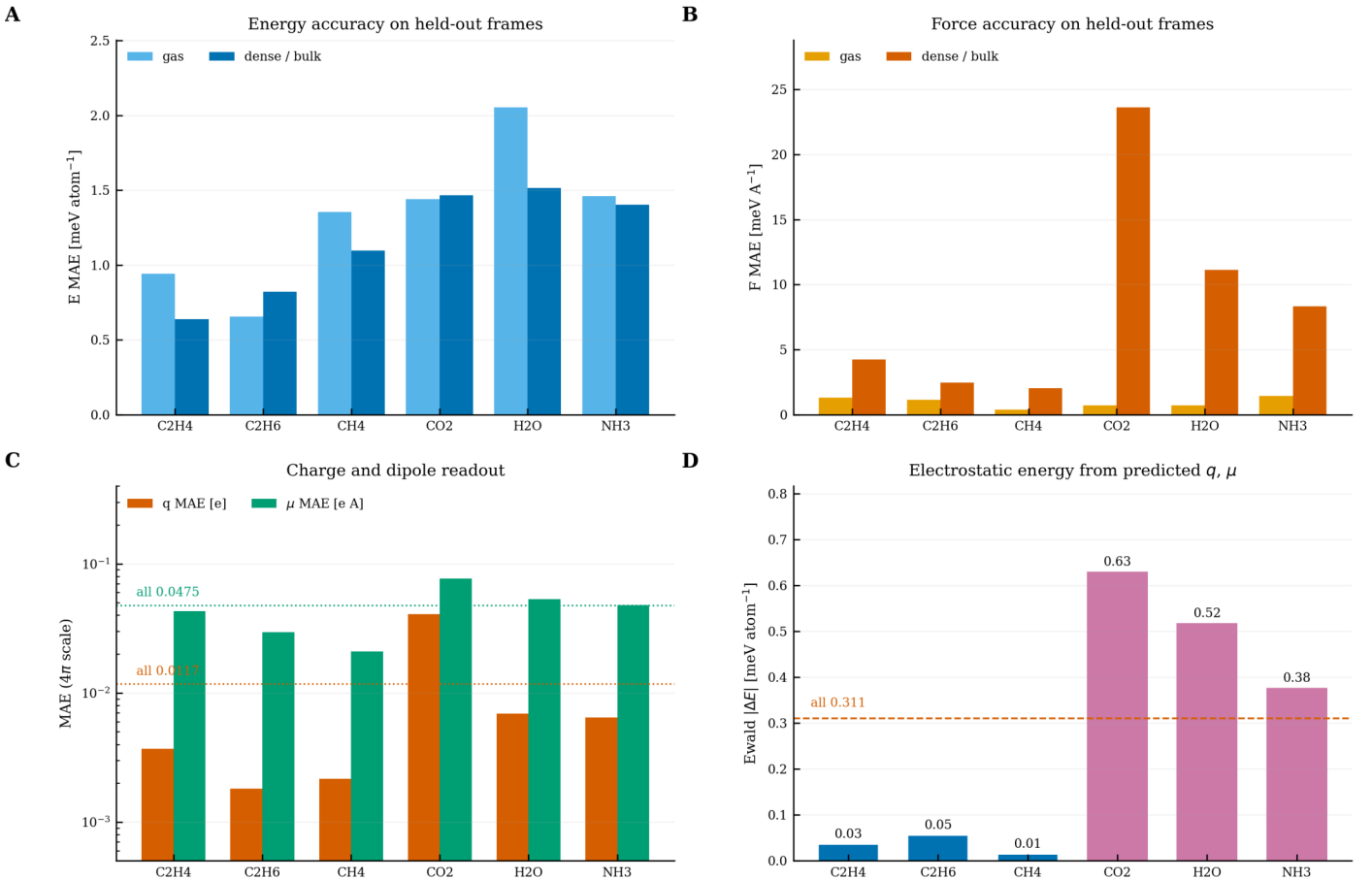
1 · Distillation architecture



Teacher distillation (*E*, F, q, *μ*), charge/dipole readout heads, and the canonical-frame + GTO/Ewald path.

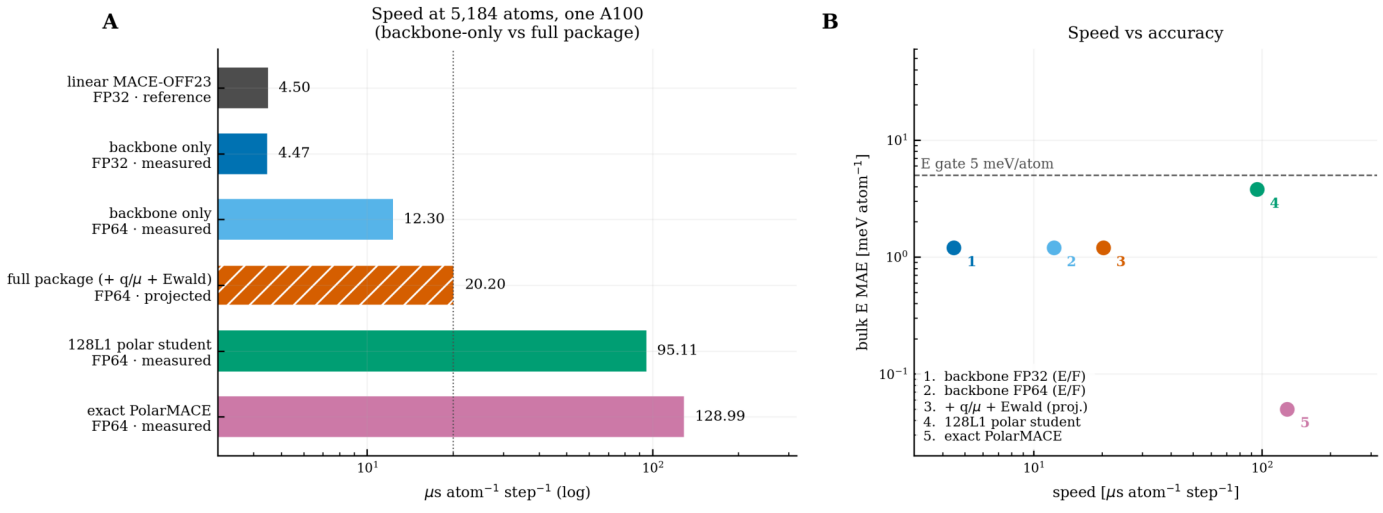
The teacher (512-channel nonlinear polar MACE, l=1 multipoles, 2-step SCF) labels a 23,376-frame E/F corpus and a 6,809-frame q/mu corpus over 1,100 states. The student backbone is a standard 128-channel linear MACE (2 interactions, r_c = 5 A, 848,528 parameters). Charges come from a 0e MLP; the l=1 multipoles come from a full linear 384->3 map in the canonical frame. The GTO/Ewald kernel is the teacher's exact long-range operator.

2 · Accuracy on the six molecules



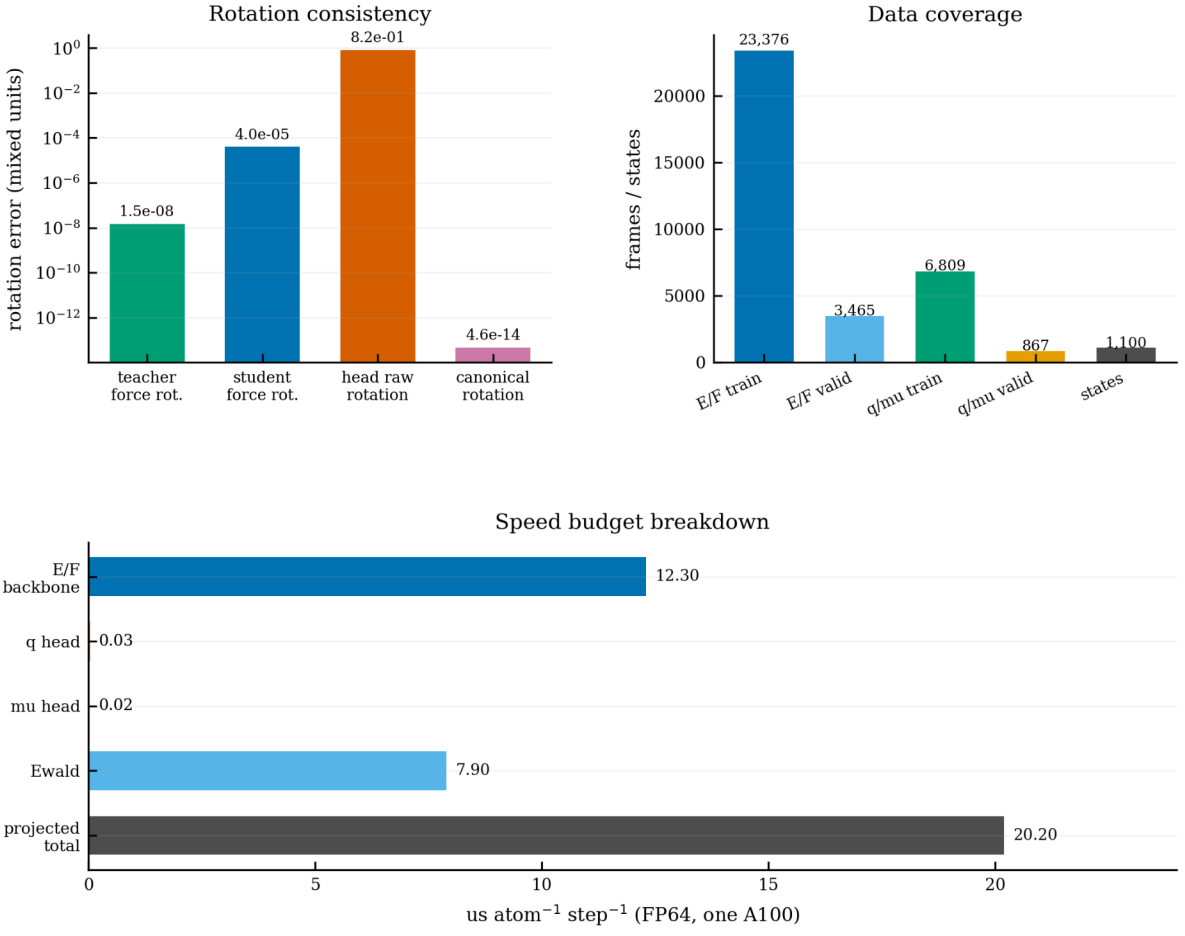
A/B: energy and force MAE of the distilled backbone on the 3,465 held-out frames, split into gas and dense/bulk states for each molecule. C: charge and l=1 dipole MAE (4pi scale) on the 867 frames that carry teacher multipoles; dotted lines are the all-frame values. D: electrostatic energy computed from the predicted q and mu with the teacher GTO/Ewald kernel, compared with the same kernel on teacher coefficients. CO2, H2O, and NH3 carry the residual (purple bars); the representation, not the frame count, is the limit. CO2's larger absolute force error tracks its ~2x larger reference forces (1.34 vs 0.65-0.76 eV/A): its relative error is 1.8%, similar to H2O (1.7%) and NH3 (1.3%).

3 · Speed and the speed-accuracy map



All measurements are us/atom/step at 5,184 atoms on one A100. The FP32 direct/JIT engine path runs the bare backbone at 4.47 (no q/mu heads, no Ewald), matching the paper's linear MACE-OFF23 range; the same backbone in FP64 is 12.3. The full package adds the q/mu heads and the exact GTO/Ewald kernel and is projected to land at ~20.2 in FP64 (hatched), because the engine hybrid pair style is not yet implemented. The 4.47 and the ~20.2 are therefore different scopes, not two measurements of the same quantity. The 128L1 nonlinear polar student (95.1) and the exact teacher (129.0) bracket the current exact-PolarMACE cost.

4 · Stability, data coverage, and the speed budget



The student force field rotates to 4.0e-5 and the NVE drift is 0.245 meV/atom/ps over 1 ps. The raw full linear head is not rotation-consistent (0.825 meV/atom); the canonical-frame transform fixes it to 4.6e-14 without retraining.

Data coverage: 23,376 / 3,465 E/F train/valid frames and 6,809 / 867 q/mu frames over 1,100 states. The ACF analysis gives ~4,450 effective independent samples, so frame counts are not independent-sample counts.

The speed budget sums the measured engine backbone (12.3 FP64), the measured PyTorch Ewald kernel (7.9 at 1,152 atoms) and the measured heads (<0.1); the 20.2 total is a projection, not an end-to-end engine measurement.

Theory and methods

Teacher and data

The teacher is MACE-Polar-1-M: a 512-channel nonlinear polar MACE with l=1 atomic multipoles, exact GTO/Ewald long-range electrostatics, and two charge-density recursion steps. The distillation corpus contains 23,376 training and 3,465 validation frames of six molecules (H2O, NH3, CH4, C2H4, C2H6, CO2) across 1,100/176 states, with a state-level split. The ACF analysis gives about 4,450 effective independent samples for the full 26,841-frame corpus; the 23K frame count is not a 23K independent-sample count.

Backbone

The student is a standard linear MACE: 128 channels, two interactions, l_max=3, r_cut=5.0 A, 848,528 parameters. It is trained on the teacher's total energy and forces (REF_energy/REF_forces) for 60 epochs, force weight 20, Adam with ReduceLROnPlateau. The model is exported to the Symmetrix JSON contract and runs on the FP32 direct/JIT engine path at 4.47 us/atom/step (5,184 atoms); the FP64 materialized reference is 12.3 us.

Charge and multipole readout

The 6,809-frame q/mu set is the stride-4 train set plus a 965-frame state supplement, covering all 1,100 states. The 0e and 1o node features are extracted at the energy readout. A two-layer MLP maps 0e features to per-atom charge; a full linear map (384->3) maps the 1o features to the teacher's three l=1 multipole coefficients. The linear map is not a per-channel equivariant map: the teacher's coefficients come from the nonlinear charge-density head and require channel/basis mixing. The physical energy and forces remain equivariant because the GTO/Ewald kernel is covariant.

Canonical frame

The full linear head is orientation-dependent in the raw frame. A 3x3 cell QR rotates every frame to a canonical (upper-triangular) orientation before the head and the Ewald kernel, and the outputs are rotated back. The transform costs one 3x3 QR per frame and restores rotation consistency from 0.825 meV/atom to 4.6e-14 meV/atom. The current orthorhombic training cells are already canonical, so no retraining is required; triclinic cells would need canonicalized retraining.

Ewald diagnostic

The predicted q and mu are fed to the same graph_longrange GTOElectrostaticEnergy kernel used by the teacher (l=1, 1.5 A smearing, k-space factor 1.5, self and PBC corrections). On 867 held-out frames the predicted-coefficient electrostatic energy matches the teacher-coefficient energy to 0.311 meV/atom overall.

Important bookkeeping

The MACE backbone was trained on the teacher's total energy, which already contains the long-range term. The q/mu head and E_elec are output/diagnostic quantities; E_elec must not be added to the MACE total energy unless the backbone is retrained on the short-range residual. This distinction is what makes the current package physically consistent without double counting.

Design decisions, improvements, and open items

Decisions

1. The 4G-HDNNP accuracy line was closed: wide+ee4G remains at 31.0 meV/atom and 75.1 meV/A (6.2x/2.5x over the 5/30 gate), active learning moves it by ~1%, and the overfit probe floors the force error near 60 meV/A. The missing l=1 multipoles and thin angular basis, not the data volume, are the limit.

2. The linear MACE distillation is the E/F main line: 1.2/7.8 on the full 23K corpus, 1.17/1.00 on the six-molecule gas set, and 4.47 us/atom FP32 on the engine.

3. The q/mu head is post-hoc on frozen features. The strict per-channel equivariant linear head is limited to ~0.227; the full linear map reaches 0.045 but is not rotation-consistent. The canonical frame resolves the rotation issue without a tensor product or a slower equivariant network.

4. Tensor-product tests: a single bilinear 1o x 1o layer is exactly equivariant (3.6e-6) but only reaches mu MAE 1.004; 2e x 1o reaches 0.905 at 1.5 us/atom; 0e+1o+2e reaches 0.905 at 3.0 us/atom. A deep equivariant network would be needed for 0.045 but would exceed the 20 us budget. Ewald is the preferred long-range backend: 7.9 us/atom median vs 9.3 for PME at 1,152 atoms, and it is the teacher's exact kernel.

Open items

- The 20.2 us total is a projection (engine backbone 12.3 + PyTorch Ewald 7.9 + heads); the Symmetrix standard MACE pair style has no charge/dipole output. An engine hybrid (standard MACE + q/mu readout + Ewald) is the remaining implementation.

- q/mu labels cover 6,809 frames, not 23K. The remaining 16,567 frames are mostly correlated frames inside the same 1,100 states; labeling them is optional for accuracy and mainly a consistency choice.

- CO2/H2O/NH3 retain 0.4-0.6 meV/atom Ewald error; 0.1 meV/atom is not reached. This is a representation (multipole/local-basis) limit, not a frame-count limit.

- The site-field (JCTC-style four-sine scalar potential) and inhomogeneous-density tranches are being sampled separately; the MACE-Polar teacher remains the label source. Electric-field response requires the MACE-Polar external-field block, not MACE-OFF23.

Reproducibility

All scripts and JSON diagnostics are in the repository: scripts/charge_dipole_student.py, eval_full_valid.py, test_package_forces_nve.py, and docs/results/data/. Model hashes are recorded in the 500 ps audit package and in the Plan D log. Figures are generated by docs/results/distill_figures.py.